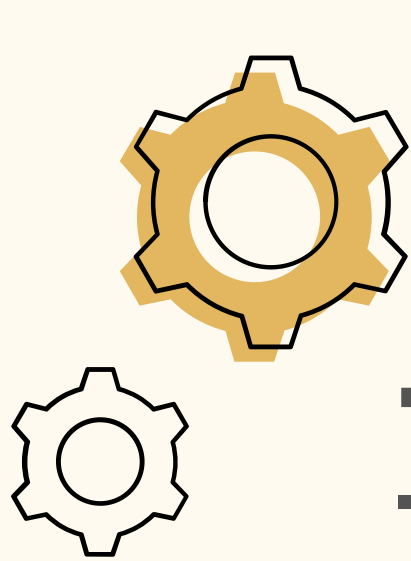


TATA-Data Visualisation: Empowering Business with Effective Insights

By Ankita Sarkar





TASK-01

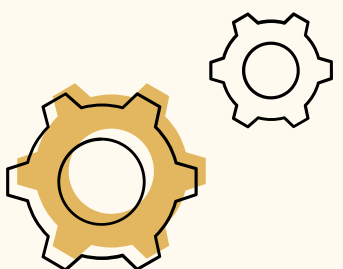
To prepare for your meeting, you need to draft questions that you think will be important and relevant to the CEO and CMO. This preparation will be your guide as you develop your presentation.

For this task, you are only required to draft the questions. Make sure to think both quantitatively and qualitatively.

You've been provided a dataset in the resources below to use as the basis for your exploration. Review this data, taking note of what information has been provided, what insights you can garner, and what is relevant to both the CEO and CMO respectively.

Create a set of four questions that you anticipate each business leader will ask and want to know the answers to. Make sure you differentiate your questions, as both the CEO and CMO view business decisions through different lenses.

Submit your eight questions in total (4 for the CEO and 4 for the CMO) in the text submission box below.



Questions for the CEO



1. Revenue Growth & Monthly Trajectory:

"What is the monthly revenue and order quantity trend across the entire dataset, and are there specific seasonality patterns or key operational milestones driving top-line growth?"

2. Geographic Revenue Distribution & Expansion:

"Which geographic markets generate the highest total revenue outside of our primary market (UK) (91.43% of total transactions (495,478 out of 541,909 rows) originate from the United Kingdom), and where should we allocate capital for international operational expansion?"

3. Customer Concentration & Revenue Risk:

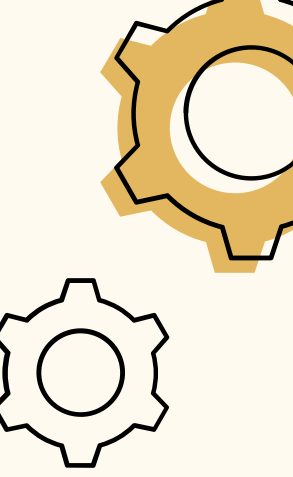
"What proportion of overall revenue is generated by our top 10% highest-value customers, and how vulnerable is the business to customer churn among these key accounts?"

4. Return Rates & Operational Efficiency:

"What percentage of total orders/items consist of order cancellations or negative-quantity returns, and what is the estimated bottom-line financial impact of these operational inefficiencies?"



Questions for the CMO



1. Customer Purchasing Frequency & Lifetime Behavior:

"What is the average order frequency and purchasing cadence per customer, and how many repeat buyers vs. single-purchase buyers do we currently have in our database?"

2. Top-Selling Products & Cross-Selling Opportunities:

"Which top 10 products generate the highest sales volume, and which complementary products are most frequently purchased together in single transactions?"

3. Geographic Marketing Targeting:

"Which international countries have the highest average order value (AOV), and where should marketing spend be prioritized to maximize return on ad spend (ROAS)?"

4. Seasonal Marketing & Repeat Engagement:

"At what specific times of day and months of the year do overall transaction volumes peak, and how can marketing run targeted re-engagement campaigns to capitalize on these peak purchase windows?"

Here is your task-02

In this task, you will create the visuals around four of the questions that the CEO and CMO have requested. You can use either Tableau or Power BI to create the visuals and we've provided the data in the resources below.

Both Power BI and Tableau have free trials that can be accessed in order to complete this task. You can find a resource below that directs you how to download these free trials.

Data cleanup

Before you can begin the analysis, make sure that the data is cleaned properly. You have noticed that the data contains some returns to the store which are provided in negative quantities and there are unit prices which were input in error. You will need to perform the following steps to clean this data.

Create a check that the quantity should not be below 1 unit

Create a check that the Unit price should not be below \$0

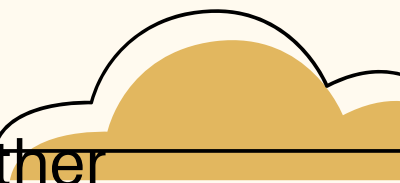
Please note that in order to apply the checks that have been mentioned above, you would need to use conditional formulas where the logic would state that if the conditions are met then the tool should exclude the data from analysis. You can also use data transformation methods to get rid of the bad data. Both these methods are provided in the resources. Once this is done, the data will be good to be used for further analysis. Please note that this data should be cleaned up before attempting any question.

Create each visual on a separate tab in your excel spreadsheet with the name of the tab displaying the question number. Once the visuals are created, save the files and upload either your work either as a .csv spreadsheet or a data file below. If you are more comfortable working straight from a .csv file, one has been provided for you.

For Tableau users, you would need to upload in .twbx format whereas Power BI files need to be saved in .pbix format.

Instructions on how to save in these formats are provided in the resources below.

Should you get stuck, please reference the Visuals Hints document in the resources.



Question 1

The CEO of the retail store is interested to view the time series of the revenue data for the year 2011 only. He would like to view granular data by looking into revenue for each month. The CEO is interested in viewing the seasonal trends and wants to dig deeper into why these trends occur. This analysis will be helpful for the CEO to forecast for the next year.



Question 2

The CMO is interested in viewing the top 10 countries which are generating the highest revenue. Additionally, the CMO is also interested in viewing the quantity sold along with the revenue generated. The CMO does not want to have the United Kingdom in this visual.

Question 3

The CMO of the online retail store wants to view the information on the top 10 customers by revenue. He is interested in a visual that shows the greatest revenue generating customer at the start and gradually declines to the lower revenue generating customers. The CMO wants to target the higher revenue generating customers and ensure that they remain satisfied with their products.

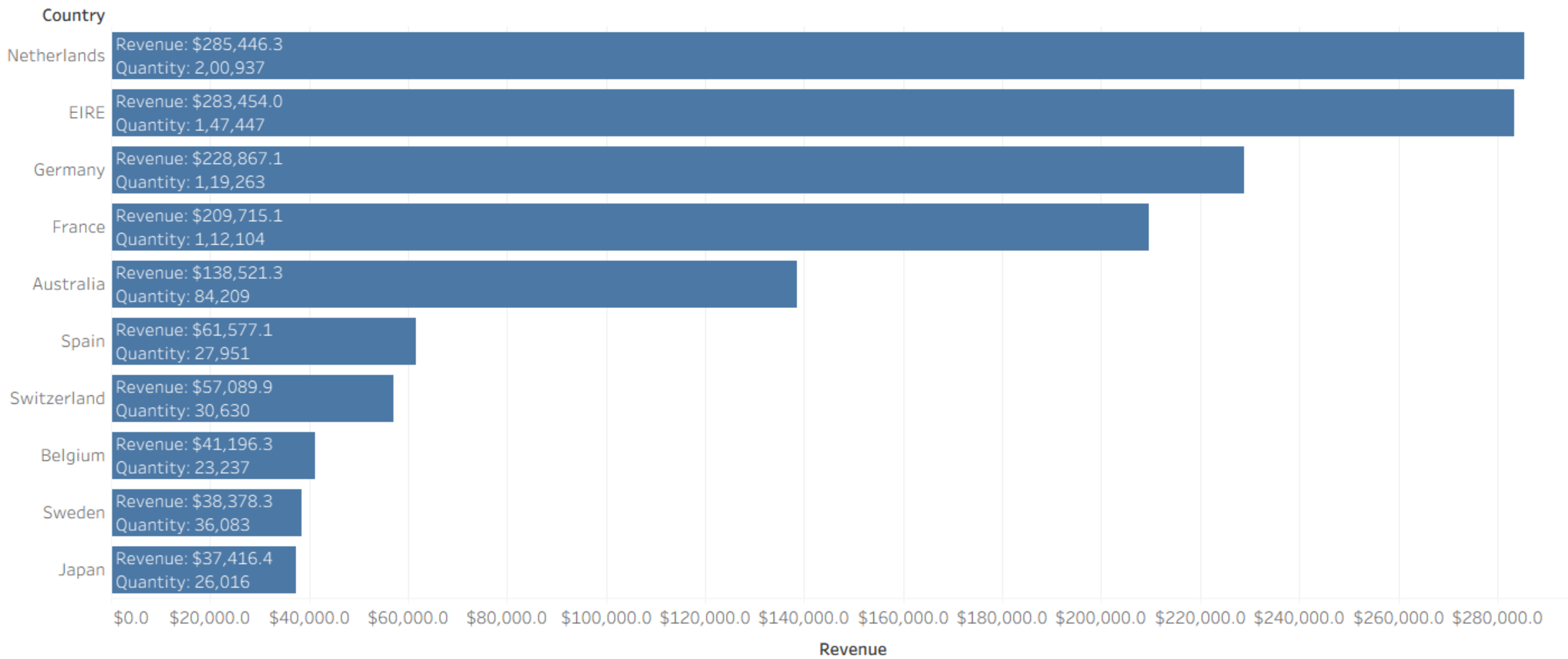
Question 4

The CEO is looking to gain insights on the demand for their products. He wants to look at all countries and see which regions have the greatest demand for their products. Once the CEO gets an idea of the regions that have high demand, he will initiate an expansion strategy which will allow the company to target these areas and generate more business from these regions. He wants to view the entire data on a single view without the need to scroll or hover over the data points to identify the demand. There is no need to show data for the United Kingdom as the CEO is more interested in viewing the countries that have expansion opportunities.

Q1 – Monthly Revenue 2011

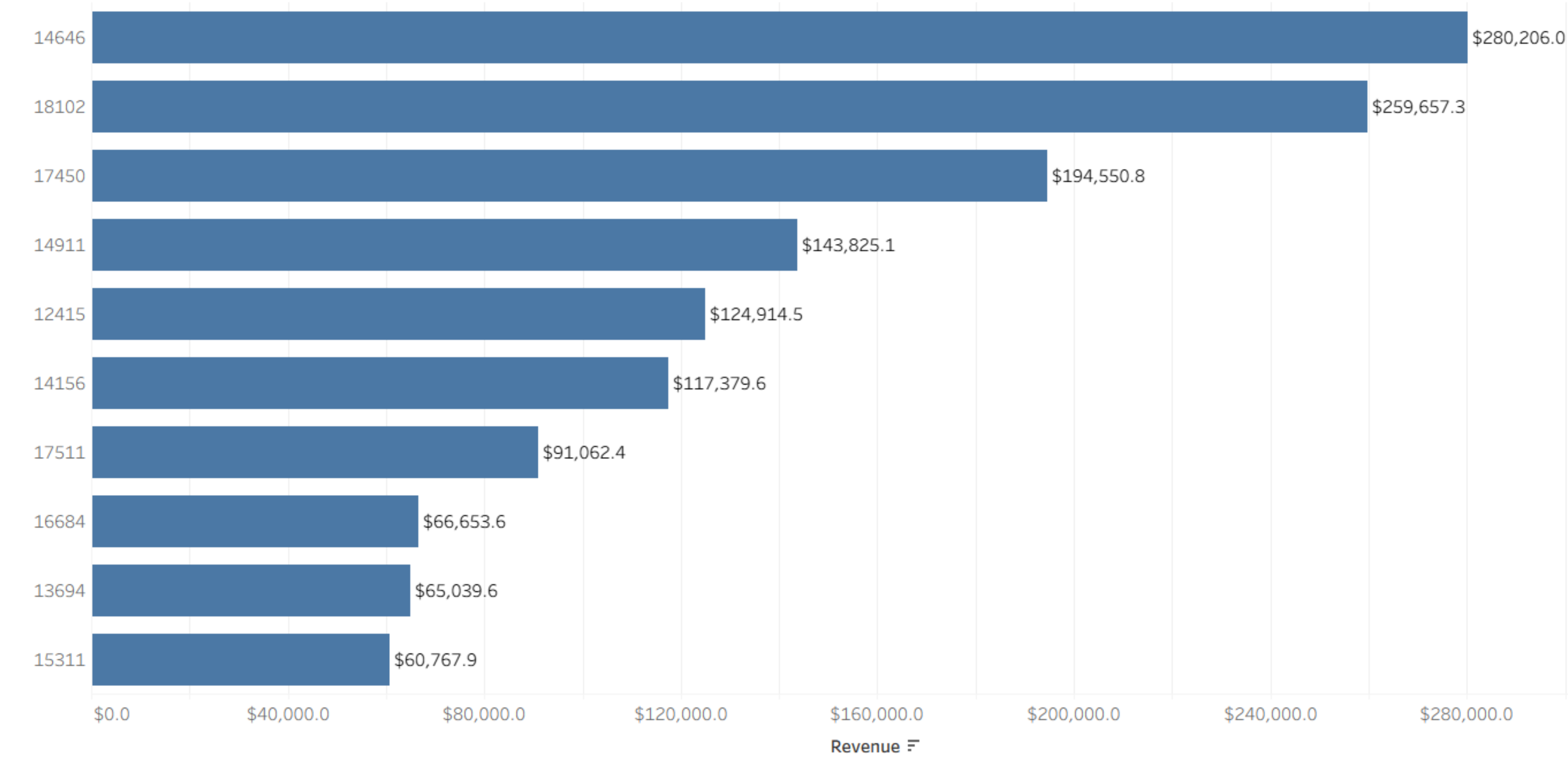


Q2 – Top 10 Countries

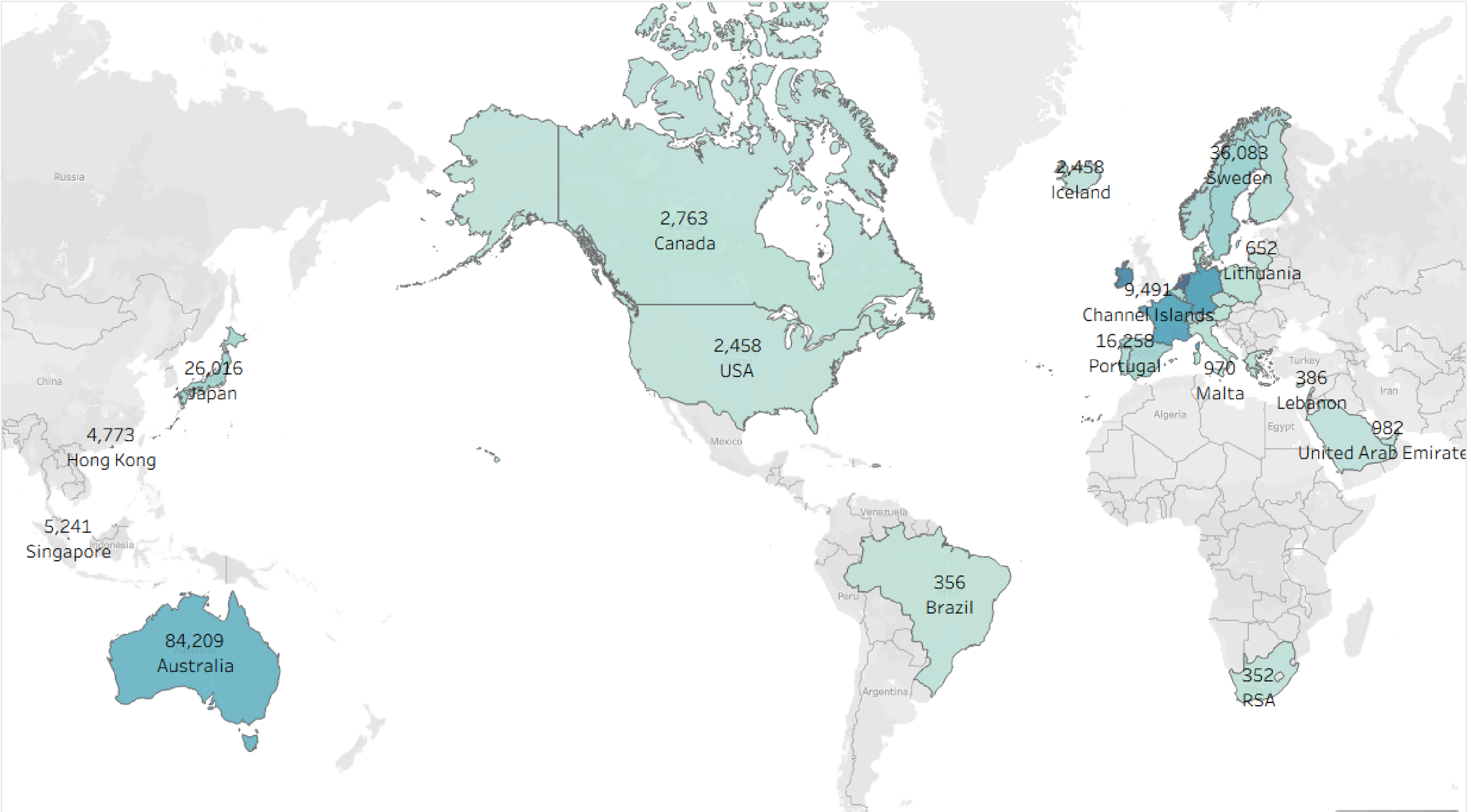


Q3 – Top 10 Customers

Cust.. #



Q4 – Global Demand



Conclusion

- To conclude, the analysis highlights four key findings.
- First, 2011 revenue shows a strong increase toward the final quarter, with November being the peak month.
- Second, outside the UK, the Netherlands, Ireland, Germany and France are among the strongest revenue-generating markets in the visual.
- Third, a relatively small group of high-value customers contributes substantial revenue, making customer retention an important consideration.
- And finally, the global demand analysis highlights markets such as Australia, Sweden and Japan where product demand is visible outside the UK.

These insights can support decisions around seasonal planning, international marketing, customer retention and potential market expansion.

Thank You

~Ankita Sarkar

